KLH UNIVERSITY
BACHUPALLY

Department of
Computer Science & Information Technology

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PUBLIC TRANSIT PLANNER

Plan • Track • Compare • Commute Smarter

A unified platform for smarter, easier and more sustainable urban commuting.

PROBLEM STATEMENT

Urban commuters face **fragmented transit information**, scattered schedules, unpredictable delays, and no unified way to compare routes across different modes of transport.

OUR SOLUTION

A digital platform that helps users **plan, track, and optimize** their daily commute across buses, metros and other public transit options in a single interface.

KEY FEATURES

Route Optimization
Fastest, cheapest or least-transfer route

Real-time Tracking
Live vehicle tracking and arrival predictions

Fare Estimation
Get estimated fare for different routes

Multi-Modal Compare
Compare bus, metro and other options

Offline Access
Access schedules even without internet

Accessibility Aware
Routes for differently abled commuters

SYSTEM ARCHITECTURE

User → **Frontend** (Web / Mobile App) ↔ **Backend Server** (Route Planning, Tracking & Fare Estimation) ↔ **Transit Data Sources**

Transit Data Sources:
- Bus Data
- Metro Data
- Schedules
- Real-time Tracking

ALGORITHMS & DSA USED

Graph Representation
Stations/locations as vertices and connections as edges

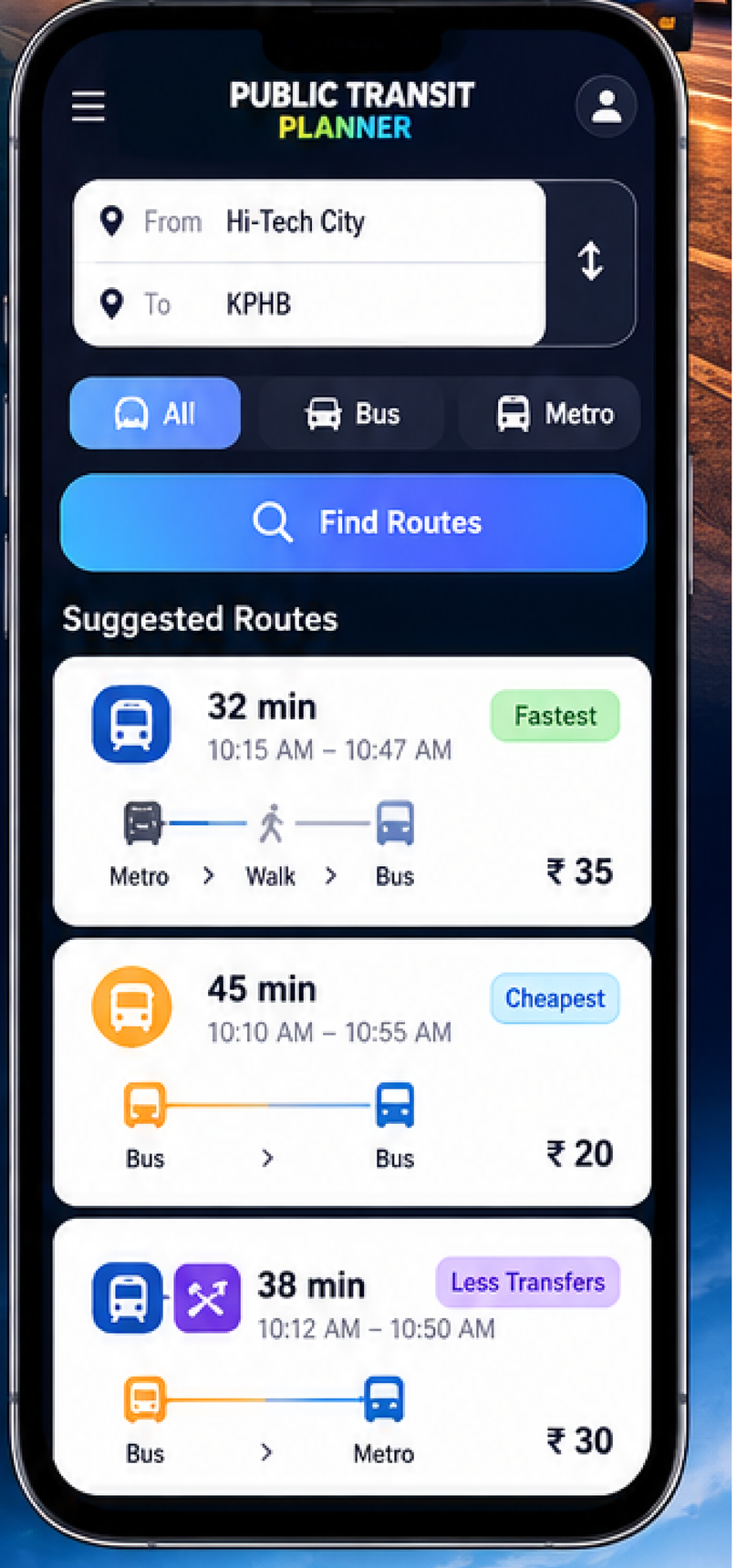
Shortest Path Algorithms
Find optimal routes based on time, fare or transfers (e.g., Dijkstra's).

Data Structures
Efficient storage and processing of transit data (Arrays, Lists, Maps, etc.).

Optimization Techniques
Compare multiple routes and choose the best based on user preference.

IMPACT & BENEFITS

- Reduces commute uncertainty
- Encourages public transport adoption
- Supports sustainable urban mobility
- Scalable across cities with varying transit infrastructure





Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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